

Michael Papamichael

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[LinkedIn](#) • [GitHub](#)

Work Permit: UK Passport & EU Citizen

Professional Summary

Quality assurance engineer and MSc graduate (Imperial College London) with 3+ years working at the intersection of financial data systems, statistical analysis, and software engineering. Experienced in Python, SQL, and applied machine learning, with hands-on delivery of predictive models, automated data pipelines, and large-scale financial dataset validation across FCA-regulated environments. Now focusing fully on data science and ML, bringing both the technical depth and financial domain expertise to deliver immediate impact.

Work Experience

Investment Software LTD | QA Engineer (Data, Automation & Technical Lead)

09/2022 – Present | London, UK

- Designed and executed complex SQL pipelines (CTEs, joins, query optimisation) to validate, map, and migrate AUM datasets scaling to billions of pounds across wealth management client onboardings, ensuring zero FCA CASS compliance failures prior to platform ingestion.
- Designed a 100K+ datapoint performance analytics framework that runs as an automated daily Power BI pipeline continuously monitoring system execution trends on bottlenecks and directly informing optimisations across releases.
- Built and maintained automated testing frameworks in C# with Selenium across UI and API layers, integrated into Azure DevOps CI/CD pipelines, ensuring data integrity and calculation consistency across development, UAT, and production.
- Technical lead on the company's new client-facing mobile application, holding full ownership of QA strategy, architecture, build, deployment, versioning decisions and security across the full development lifecycle.
- Led QA design and delivery of the company's back-office module from inception to client deployment, driving weekly stakeholder requirement sessions and ensuring scalability and robustness throughout.

H2Go Power LTD | Data Scientist (Capstone Project during MSc)

06/2022 – 08/2022 | London, UK

- Developed an end-to-end ML pipeline utilising ARIMA time series forecasting to model raw material and logistics costs over a 10-year horizon, delivering the company's first quantitative supply chain planning framework to inform strategic investment decisions.
- Conducted comprehensive data acquisition and EDA, extracting market intelligence from multiple sources, performing feature engineering, and uncovering critical supply chain dependencies and cost drivers that underpinned the forecasting model.
- Delivered an executive Tableau dashboard communicating predictive findings, statistical findings and scenario analysis directly to the CTO, shaping strategic investment and supply chain decisions.

Cyprus National Guard - Military Service | Master Sergeant

07/2017 – 09/2018 | Cyprus

- Advanced to supervisory rank via competitive assessment, managing a 20-person team across operations, shift planning, and training, improving readiness by 10%.

Personal Projects

UK Fuel Pricing Fairness Model | [Live Demo](#) | [GitHub](#)

- Building an end-to-end overcharging detection model for UK petrol stations using the government's Fuel Finder open dataset: predicting the fair price each station should charge given wholesale costs, brand, and local competition, then flagging stations with large positive residuals.
- Engineered an automated data collection pipeline in Python using the Fuel Finder OAuth2 API, scheduling daily ingestion into date-partitioned Parquet queried with DuckDB SQL, and deployed a live Streamlit app early on to validate the pipeline.
- Engineering spatial competition features with GeoPandas, a LightGBM fair-price regression validated with spatial cross-validation and SHAP explanations, then extending the analysis with an asymmetric pass-through model in statsmodels.

NY Real Estate Price Prediction | [Live Demo](#) | [GitHub](#)

- Built a ML pricing model using XGBoost and NLP feature engineering (TF-IDF, embeddings) on a messy dataset (77% missing values, KNN imputation) to predict NYC property prices from 8,000+ listings, achieving an R^2 of 0.742.

Education

Imperial College Business School | MSc Business Analytics (Applied AI/ML) (Merit) | 2021 – 2022 | London, UK

Relevant Modules: Advanced Machine Learning, Financial Analytics, Statistics & Econometrics, Data Structures & Algorithms, Data Visualisation

University of Manchester | BEng Mechanical Engineering (2:1) | 2018 – 2021 | Manchester, UK

Individual Project: Built an ML pipeline in Python integrating speech recognition models with Fortran-based engineering simulation software on Linux. The system used custom-trained speech recognition models to enable voice-controlled automation of simulation workflows, an early application of NLP before the current wave of AI tooling.

Languages & Skills

Languages: English (Native), Greek (Native), French (Conversational)

Programming & Tools: Python, SQL, C#, Power BI (DAX, semantic modelling), Tableau, Azure DevOps, CI/CD

ML & Data Science: Regression, Classification, Time Series Forecasting, Clustering, XGBoost, NLP (TF-IDF, Sentence Embeddings), PCA, Regularisation, Feature Engineering, EDA

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Selenium